

Daily Science Anecdote

Topic: You Become Spider-Man?

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Have you ever watched the Spider-Man movies and then in your sleep, you dreamed about becoming Spider-Man? What if I tell you that it is not a dream anymore, almost.

So you have encountered a spider web in the basement and it felt so easy to cut, that's because the fibers which are made of spider-silk are so thin. A spider web is made up of seven different types of spider-silk - each one serving a purpose. If we get more of the strongest type of spider-silk then form a thick rope, it is proved to be stronger than steel and tougher than Kevlar (Kevlar is a material used to make military armor). This strong spider-silk is used by spiders in webs that are 25 meters wide across rivers in the forests. But since the spider-silks are in the deep forests, it costs $50 \times \text{gold}$ which is almost 7 million dollars per Kilogram of spider-silk. If you want to make it yourself to become spider-man, unfortunately we still do not fully understand how it is made by spiders, but what if the spiders make it for us in laboratories? As the spiders are in the lab, the only thing left is to make enough of it to make a rope. But spiders can't produce spider-silk enough or it would take too long. However the silk you are familiar with comes from silkworms and is used in clothing, so why not the silk-worms make it for us since they can make a lot of it? Here we move DNA from spiders to silk-worms so they produce the strong spider-silk in large amounts that we need. Derek from Veritasium got a sample of this spider-silk rope in his hands and... He hung onto it in the air. He even started swinging high in the air although it was so strong and thin it cut into his hands. He officially became the first spider-man!

So when can you become spider-man? Not today or tomorrow, but one day you will be able to wear spider-silk clothing or products that use its strength and other useful properties. It is fantastic to see that the smallest things like spiders can produce the strongest materials, making clothing stronger, enhancing armor or even helping in nerve surgery as its so small yet robust.

Next time you come across a spider web, just know that when stacked, it can withstand bullets.